# A Four-Dimensional Model of Hydro Generation for the Short-Term Hydrothermal Dispatch Problem Considering Head and Spillage Effects

Andre Luiz Diniz, *Member, IEEE*, and Maria Elvira Piñeiro Maceira

***Abstract*—The generation of a hydroelectric plant is a nonlinear function of the turbined outflow, storage and sometimes also of the spillage. Other aspects, such as forbidden operating zones and individual turbine efficiency curves, are also important. Although for the self-scheduling of a hydro plant a precise integer modeling of the hydro power production function (HPF) is convenient, in the security constrained short-term hydrothermal dispatch problem for large-scale systems a strategy which best balances an accurate representation with an acceptable computational burden is required.**

**In this paper, a detailed review of different approaches to model the HPF is presented, and a new four-dimensional piecewise linear model is proposed. This model takes into account the water head as a function of forebay and tailrace levels and considers spillage effects. Nonconcave regions of the HPF are approximated by using convexification and regression techniques. A major advantage of this approach is to consider the influence of storage, turbined outflow and spillage in a single function that can be used in a straightforward way, instead of deriving several curves for different heads. Numerical results show the applicability of this approach to the day-ahead network constrained short-term hydrothermal dispatch problem of the large Brazilian system.**

***Index Terms*—Hydroelectric power generation, linear programming, power generation scheduling.**

## I. Introduction

In the operation planning of hydrothermal power systems, it is crucial to have a suitable modeling of the hydro generation as a function of the turbined outflow and the head in the reservoir—that in turn depends on storage, turbined outflow and sometimes also on spillage. The accuracy of the representation usually depends on the problem addressed.

For the long term planning, it is quite usual to consider an equivalent reservoir production function, obtained by aggregation of several plants in a single hydro valley [1], [2].

At the opposite side, in the self scheduling [3], [4], or the hydro unit commitment problem [5], [6], hydro plants are to be dispatched for the next day given predicted system marginal costs or daily generation/discharge targets. In these situations, each hydro unit of a plant is usually considered individually, taking into account their efficiency curves [5], [7]—called hill curves or hill diagrams—and forbidden zones [5], [8], by means of a detailed 0–1 and nonlinear representation.

In the centralized short-term dispatch problem for large-scale hydrothermal systems, with network security constraints [9], [10] and for an horizon of up to a week on a hourly basis, the computational burden involved in a highly sophisticated modeling of the hydro units is very cumbersome. Instead, it can be useful to build a hydro generation model that suitably manages the trade-off between an accurate representation of water head variation and low computational times.

In this paper, we propose a four-dimensional piecewise linear model for the generation of a hydro plant, as a function of storage $(V)$, turbined $(Q)$ and spilled $(S)$ outflows. The procedure is done in two steps. We first calculate the hyperplanes that define the convex hull of the region below the analytic hydro production function (HPF) in the $V \times Q$ plane (considering $S = 0$), to obtain an initial 3-D piecewise linear approximation for the HPF. To avoid optimistic approximations, a minimum average square error algorithm is applied to adjust the hyperplanes. Finally, secant approximations are computed in the spillage axis.

Two major advantages of this approach are: the detailed representation of spillage and the consideration of head variation in a single multivariate linear function of turbined outflow, storage, and spillage. Numerical application to the day-ahead network constrained short-term hydrothermal dispatch problem for the large Brazilian system shows quite acceptable computational times and small relative errors for the generation of the hydro plants, when compared to the real generation function.

The rest of the paper is organized as follows. The next section describes the HPF. Section III presents a review of several approaches to model this function. Section IV details the model proposed, Section V presents the numerical results and the last section contains the conclusions of this work.

## II. Hydro Power Generation

The power of a hydro unit $(gh)$ depends on the turbined outflow of the unit $(q)$, its design characteristics (represented by both turbine $(\eta_t)$ and generation $(\eta_g)$ efficiency factors) and the net water head in the reservoir $(h)$, as stated in the following:

$$gh = 9.81 \times 10^{-3} \eta_t \eta_g q h \tag{1}$$

where the numerical factor takes into account the gravity acceleration, water density and makes the appropriate unit conversions from (m) and $(\mathrm{m}^3/\mathrm{s})$ to MW.

Manuscript received April 30, 2007; revised January 27, 2008. Paper no. TPWRS-00299-2007.

The authors are with CEPEL, Brazilian Electrical Power Research Center, Rio de Janeiro, Brazil (e-mail: diniz@cepel.br; elvira@cepel.br).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TPWRS.2008.922253

0885-8950/$25.00 © 2008 IEEE
Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

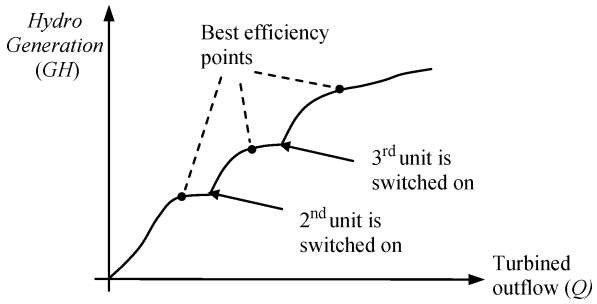



Fig. 1. Typical I/O function for a hydro plant with three units.

The net head in the reservoir is itself given by

$$h = h_{up}(V) - h_{dw}(Q, S) - h_{loss} \tag{2}$$

where $h_{up}$ is the forebay level, which depends on the volume in the reservoir $(V)$, and $h_{dw}$ is the tailrace level, which is a function of the turbined outflow of the whole plant $(Q)$ and, depending on the plant site configuration, sometimes also of the spillage $(S)$. The last term $h_{loss}$ accounts for penstock head losses.

Turbine efficiency is expressed by the hill curve of the unit, which is a function of the net head and the turbined outflow, whose level sets are shown in [5]

$$\eta_t = \eta_t(h, q) \tag{3}$$

while the generator efficiency depends only on the power [7]

$$\eta_g = \eta_g(gh). \tag{4}$$

Summing up the outputs of each unit, the following expression for the power generation of a hydro plant with $nh$ units is obtained:

$$GH = \sum_{i=1}^{nh} gh_i(q_i, V, Q, S). \tag{5}$$

This hydro plant input/output (I/O) relationship has been referred to in the literature as: production function [5], unit performance curves [3], [11], water rate curves [6], generation characteristic [12] and water-power conversion [13]. In this work the term hydro production function (HPF) will be used.

Complicating issues in modeling a curve for the generation of the whole hydro plant are: 1) the units forbidden zones; 2) the number of running units; and 3) the sequence in which the units are turned on/off. Moreover, even when a given start-up/shutdown sequence is given, the resulting I/O curve has nonconvex regions below the curve, as well as discontinuous derivatives at those points where new units are switched on, as shown in Fig. 1.

We discuss next the main parameters and variables of the HPF.

### A. Unit Efficiency Factor

The unit efficiency factor is the product between the turbine and the generator efficiency factors. It should be clear that this efficiency is not the head effect on the produced power, which is discussed in Section II-D.

A detailed representation of the unit efficiency factor is very important in the self-scheduling problem of a hydro plant [4], or the hydro unit commitment subproblem [5], which originates in general from decomposing a larger hydrothermal dispatch problem. Nevertheless, for the centralized security constrained short-term hydrothermal dispatch problem, there are limitations for such approaches, because the computational burden is very high for large-scale systems. Therefore, a constant unit efficiency is usually considered [13].

### B. Forbidden Zones

The hydro units cannot operate in certain ranges of power, due to low efficiency levels, cavitation phenomena or excessive mechanical vibrations [5], [8]. These constraints, although not often considered in many hydrothermal dispatch algorithms, may have an important impact on the feasible generation ranges for the whole plant.

### C. Number of Running Units

The number of running units of the plant can be considered implicitly, by constructing a single curve for the production of the whole plant, where each interval of power corresponds to a certain number of switched units (see [12] and Fig. 1). Alternatively, different curves can be considered for the same power interval, one for each number of running units [4].

### D. Head Effects

This is the main aspect to be considered when modeling the HPF for the short-term dispatch problem. As stated in (2), the net head in the reservoir depends on the forebay level, the tailrace level and the penstock head losses.

*1) Forebay Level:* The daily variation of the forebay level has little effect on the production for hydro plants with large dams [7]. However, when dealing with nearly run-of-the-river plants in a one-week time horizon, neglecting the dependence between the head and the storage in the reservoir may lead to important losses of accuracy in the modeling of hydro generation.

*2) Tailrace Level:* The tailrace level in a hydro plant increases with the plant discharge. This causes a decrease in the net head and has a negative effect on the power plant generation. Therefore, the productivity of a hydro plant $(\mathrm{MW}/(\mathrm{m}^3/\mathrm{s}))$, measured as the slope in the I/O function, also decreases with the turbined outflow, leading to a concave behavior of this function.

*3) Penstock Head Losses:* Penstock head losses are usually considered as a quadratic function on the turbined outflow [7], multiplied by a parameter that depends on the design of the pipe structure that conducts water from the water intake in the reservoir to the turbines. There can be considered either individual unit penstock losses and/or common penstock losses.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

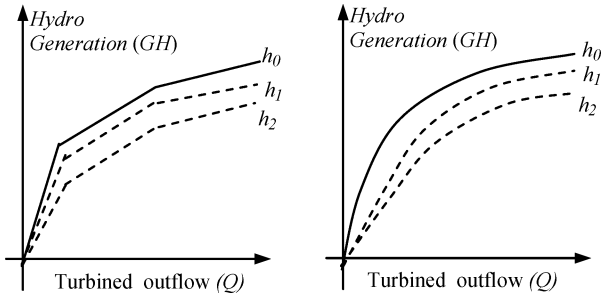



Fig. 2. Family of concave curves to model the hydro plant generation function, one for each head ($h_2 < h_1 < h_0$).

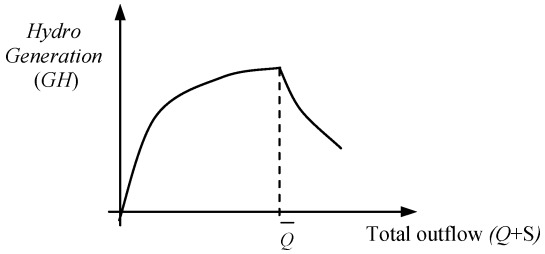



Fig. 3. Model of the HPF considering the total outflow as the independent variable.

## III. Recent Approaches for the HPF

As can be seen in Fig. 1, the region below the actual HPF of a hydro plant is nonconvex, yielding some difficulties for solving the dispatch problem with rigorous mathematical programming optimization techniques. This leads to some assumptions and concave approximations to model the HPF.

### A. Concave Approximations and Head Variation

A first representation to model the generation of a hydro plant, adequate for plants with little head effect, is to consider the generation as a piecewise linear [14], [15], or quadratic [16], [17] concave function of the turbined outflow, considering a given head in the reservoir. For head dependent reservoirs, a family of piecewise-linear [18], quadratic [19] or piecewise quadratic [11] functions can be considered, one for each given head (see Fig. 2), with appropriate interpolation for intermediate values. The use of different head curves is not restricted to the approximation being concave [3].

Although being a more precise representation, the consideration of head effect by different curves gives an iterative aspect to the dispatch problem to be solved, as the choice of the curve depends on results yet to be obtained. In order to consider the head effect in a single function, a composite representation of a linear function in $Q$ and a quadratic function in $V$ is used in [20]. In our previous work [21], a piecewise linear concave function of $V$ and $Q$ was used for the hydrothermal dispatch problem with thermal unit commitment constraints. In [22], a similar approach is considered, but nonconvex regions are explicitly modeled by introducing binary variables.

Works that use stochastic search techniques to solve the hydrothermal scheduling problem can handle easier with more precise formulations. A bi-quadratic function of both turbined outflow and storage is used in [23] and [24]. In [25], the head-dependent coefficients of a linear function of $Q$ are obtained from available tables.

### B. Best Efficiency Points

A preference may be given to operating the units at their best efficiency points, defined as the points of maximum turbine-generator efficiency. The HPF can then be modeled with a special attention to these points.

In [12], the feasible region for the HPF of a plant with $n$ units is reduced to $(n-1)$ discrete points, corresponding to the best efficiency points for each number of running units, and a linear approximation ranging from the best efficiency point with $n$ units and the maximum plant generation. Although assuring that generation occurs at one of the desired points or range of values, a drawback of this representation is that system and hydraulic constraints may force the plant to operate out of these regions. Moreover, in systems with cascaded hydro plants and several hydraulic and electrical network constraints, maximization of efficiency for each plant may not lead to the system optimal solution.

In [11], the HPF is considered linear for points below the best efficiency point and piecewise quadratic after it. In [15], the piecewise linear approximation is done by unit with the breakpoints at the minimum outflow, best efficiency and maximum outflow positions.

### C. Spillage Effects

Most of the works mentioned above do not consider the effects of spillage in the hydro plants production function. The tailrace level variation in a hydro plant is caused by the turbined outflow and, depending on the plant site configuration, also on the spillage. In the Brazilian system, huge amounts of spillage in some plants are quite common at certain periods of the year, for example in the Tucuruí hydro plant, a large plant in the North of the country. Such spilled outflows may have a high impact on the generation of the plant, and cannot be neglected.

A usual approach to consider the spillage in the HPF [26] is by changing the $Q$ variable on the x-axis for a $D = (Q + S)$ variable, which represents the total outflow of the plant. The HPF is modeled considering that spillage occurs only when the plant has reached its maximum turbined outflow $(\overline{Q})$, as shown in Fig. 3.

Although quite reasonable in practice, this assumption may not be verified if a combination of minimum outflow constraints and transmission capacity limits forces the plant to spill before reaching its turbined outflow capacity. To overcome this problem, a separate representation of spillage, in a different axis, is proposed in this work.

### D. Pumped-Storage Hydro Plants

Some hydro plants have two different operating models: generation and pumping, which cannot occur simultaneously. The typical model to deal with this situation is to extend the HPF to the third quadrant on the $Q - GH$ coordinated plane, where negative values of $Q$ and $GH$ correspond to pumped flow and load consumption, respectively (see Fig. 4). In general, discrete values are considered for pumping mode [14], [15], [19]. Another approach is to consider the pumping mode as a separate

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

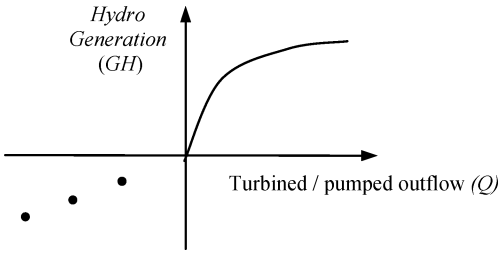



Fig. 4. Production function of a pumped-storage hydro plant.

unit, obtained from duplication of the original plant, with the appropriate adjustments in the constraints formulation.

As can be noted from this brief review, there are many different ways to model the HPF. Each approach has advantages and drawbacks, and focus on some particular aspects of hydro generation. The successful application of a given approach will depend on the problem to be solved. For instance, formulations restricted to the scheduling of a single or a small number of plants tend to allow more detailed representations. However, when dealing with the short-term dispatch problem for large-scale hydrothermal systems, considering hydraulic and electrical network constraints, a compromise solution between computation effort and suitable representation of hydro generation must be found.

## IV. Proposed Model for the HPF

We propose a 4-D piecewise linear model for the generation of the hydro plants as a function of $V$, $Q$ and $S$. The procedure is done in two steps. First, we find the hyperplanes that define the convex hull of the region below the HPF in the $V \times Q$ plane, considering $S = 0$, to obtain an initial 3-D piecewise linear approximation. A minimum average square error algorithm is then applied to adjust the hyperplanes and avoid optimistic approximations. Finally, to model the spillage effect, secant approximations are employed.

In our previous work [21], a piecewise linear model, without storage discretization and spillage effects, was applied to the hydrothermal dispatch problem considering thermal unit commitment constraints. In [27], a similar approach was applied to the stochastic midterm hydrothermal dispatch problem. Neither works consider electrical network constraints.

The main advantages of the approach proposed in this paper are: the detailed representation of the spillage, and the consideration of head variation in a single multivariate function of $V$, $Q$ and $S$. We will denote our model as AHPF—"approximated hydro production function."

### A. Initial Assumptions and Available Data

We assume a constant turbine-generator efficiency $\eta$. Penstock losses $h_{loss}$ are calculated as a percentage of the power output and we consider that they do not vary with the turbined flow. Both parameters can be computed from the known reservoir state, given at the beginning of the period of study, and from an average turbined outflow estimated *a priori*. Forbidden zones are initially neglected, and generation dispatch is calculated for each plant at each time step and further checked for feasibility of allocation among the plant units. In case of violations, new constraints are added to the dispatch problem, which is solved again.

TABLE I
Points of the HPF to be Convexified

| Point ($p$) | $Q_p$ | $V_p$ | $GH_p$ |
|---|---|---|---|
| A | 0 | 0 | 0 |
| B | 100 | 0 | 50 |
| C | 0 | 100 | 0 |
| D | 100 | 100 | 100 |

The forebay level $h_{up}$ is a fourth-degree polynomial function of the storage, and the tailrace level $h_{dw}$ is a fourth-degree polynomial function of $Q$, and in some plants, of $Q + S$. The productivity $\rho$ of each plant, defined here as the water-power conversion rate per unit of head ((MW/(m3/s))/m), is considered known. Evaporation rate at each reservoir $(\mathrm{m}^3/\mathrm{s})$ during the week is given by $c\xi A(V^0)$, where $\xi$ is a monthly evaporation factor (mm/month), $A$ is a known polynomial function that gives the surface area of the reservoir $(\mathrm{km}^2)$ as a function of its storage volume, $V^0$ is the initial storage of the reservoir, and $c$ is a unit conversion factor. Storage, turbine, spillage and generations limits are all known values.

### B. Determination of the Grid of Points for the Approximation

A grid of points in the $V \times Q$ plane is defined. The number of points and their distribution will depend on the precision required in the model and the behavior of the real plant production function. For each point $(V_i, Q_i)$, the power generation $GH_i$ considering $S = 0$ is calculated by the following formula:

$$GH = 9.81 \times 10^{-3} \rho Q \left[h_{up}(V) - h_{dw}(Q, S) - h_{loss}\right] \qquad (6)$$

to obtain a set $N$ of points $(V_i, Q_i, GH_i)$ in the 3-D space.

### C. Convex Hull Calculation

The 3-D convex hull of $N \cup M$ is calculated, where $M$ are the extreme points necessary to define a compact region below the curve $GH(V, Q)$. Determination of 3-D convex hulls is time consuming, but we have developed a customized algorithm that takes into account the particular shape of the region to be calculated.

Our procedure can be summarized as follows. For each subset $P$ of three points $(p_1, p_2, p_3)$ in the discretization grid, we find the equation $GH = a_0 + a_Q Q + a_V V$ of the plane $\widetilde{P}$ passing through $p_1$, $p_2$, and $p_3$. Then we check, for all other points $j$ in the grid, if $GH_j$ is lower than $a_0 + a_Q Q_j + a_V V_j$. If this is true for all points $j$, then $\widetilde{P}$ belongs to the convex hull of $N \cup M$. If all possible planes were considered, $C_n^3(n-3)$ verifications would be necessary. However, our algorithm takes great advantage by inspecting *a priori* the behavior of the partial derivatives of the HPF along each straight line of the grid, for a fixed value of $Q$ or $V$ (see Section IV-D).

To illustrate our high level convex hull algorithm, consider a HPF consisting of four points A, B, C and D, whose coordinates are shown in Table I.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

TABLE II
VERIFICATION IF THE PLANES BELONG TO THE CONVEX HULL OF HPF

| Plane | Points | Equation | *Point p to be checked* | *GH´(Vp, Qp) – GHp* | *Convex hull ?* |
|---|---|---|---|---|---|
| I | A,B,C | $GH' = 1/2Q$ | D | – 50 | No |
| II | A,B,D | $GH' = 1/2Q + 1/2V$ | C | + 50 | Yes |
| III | A,C,D | $GH' = Q$ | B | + 50 | Yes |
| IV | B,C,D | $GH' = Q + V/2 - 50$ | A | – 50 | No |

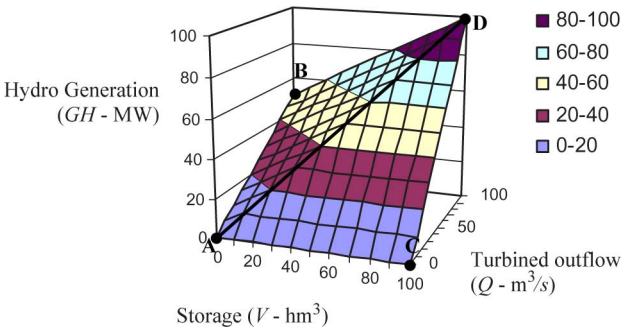



Fig. 5. Convex hull of the HPF in the illustrative example.

The four possible planes that may belong to the convex hull of the HPF are calculated and checked, as shown in Table II.

The convex hull of HPF is then composed of planes II and III, as shown in Fig. 5.

In this example, all four points belong to the convex hull of HPF. If another point $E$ with $(Q_E, V_E, GH_E,) = (50, 0, 20)$ was included in the discretization, it would not belong to the convex hull of HPF and would be eliminated.

### D. Preprocessing Algorithm

Some nonconcave regions of the HPF can be identified *a priori* wherever there is an increase in its partial derivative along any straight line. Without loss of generality, consider a set of $r$ points $(V, Q_1, gh_1), \ldots, (V, Q_r, gh_r)$ for a given value $V$, where $gh_i = GH(V, Q_i)$. For simplicity, denote $s_{i,j} := (gh - gh_i)/(Q_j - Q_i)$ as the slope of the straight line passing through points $i$ and $j$. The following lower level algorithm can then be applied.

Step 1) Initialization. $i := 1$; $s := s_{1,2}$; $I_{Ref} := 1$; $Nout := 0$

Step 2) Stopping criteria. $i := i + 1$. If $i = r$, stop.

Step 3) Derivative check. Compute $s^+ = s_{i,i+1}$.

If $s^+ \leq$ s, then $I_{Ref} := i$; $s := s^+$,

else $Nout := Nout + 1$; $Iout(Nout) := i$;

$s := s_{Iref,i+1}$;

go to Step 2.

The output of the algorithm is a set of $Nout$ points with indices $Iout(1), \ldots, Iout(Nout)$ to be eliminated from the convex hull computation. For example, in the cross section of the HPF shown in Fig. 6, points number 4, 5 and 6 would be eliminated.

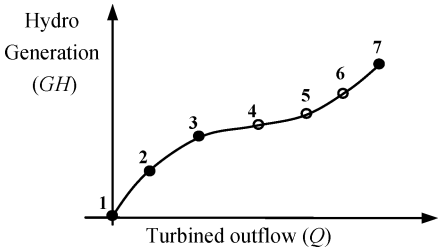



Fig. 6. Illustrative example of a set of points (marked in white) eliminated *a priori* from the calculation of the convex hull of the HPF.

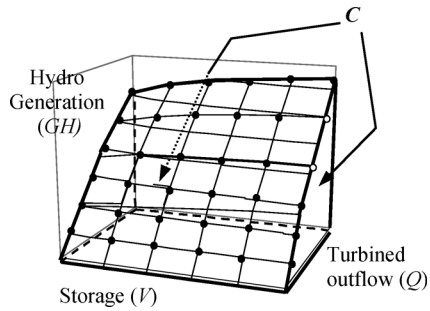



Fig. 7. Convex hull of a grid of points of the HPF.

By applying the algorithm just described for all straight lines associated to fixed values of $V$ and $Q$, it is possible to eliminate from our analysis not only several subsets $P$ of points, but also, for a given plane $\widetilde{P}$, the verification for several points in the grid.

### E. Convex Hull Results

In Fig. 7 the shape of the convexified region $C$ below a general HPF curve is illustrated.

Equations of the planes defining $C$ compose our initial AHPF model, denoted by $\mathrm{AHPF}_0$, and given by

$$GH'_i(V,Q) = \min_k \left(GH_i^{k\prime}(V,Q)\right), \quad k = 1, \ldots, K_i \qquad (7)$$

where $GH_i^{k\prime}(V,Q) = \gamma_0^{i,k} + \gamma_V^{i,k} V + \gamma_Q^{i,k} Q$, and $K_i$ is the number of hyperplanes for each plant $i$.

An important aspect that should be noted is that, if the HPF is ill conditioned (e.g., is nonconcave) in a major part of the $V \times Q$ grid, increasing the number of points in the discretization does not result in a better model, since all points in the nonconcave part of the function will be eliminated by the convex hull calculation.

### F. Adjustment of the AHPF in the $V \times Q$ Plane

The $\mathrm{AHPF}_0$ model is, by construction, always an optimistic approximation of the HPF. A more realistic representation is obtained by applying a correction factor $\alpha$ to $\mathrm{AHPF}_0$ that minimizes the average square deviation from the HPF. This is done in a second step, considering again only the $V \times Q$ plane, with $S = 0$

$$\mathrm{AHPF}(V,Q) = \alpha \mathrm{AHPF}_0(V,Q) \qquad (8)$$

which leads to $GH_i^{k\prime} \leftarrow \alpha GH_i^{k\prime}$.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

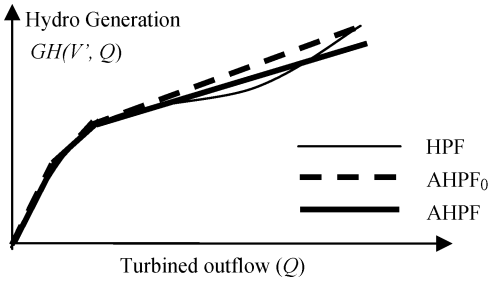



Fig. 8. Cross-section illustration of the mean-square error adjustment of the AHPF model, in the $V \times Q$ grid, for a given $V'$.

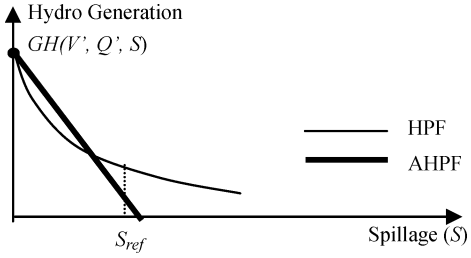



Fig. 9. Model of the HPF considering spillage as a separate variable.

This procedure is illustrated in Fig. 8 for a $GH \times Q$ cross-section of the HPF, for a certain value $V = V'$.

### G. Modeling of the Spillage

Analysis of several hydro plants showed that the original HPF has a convex shape in the dimension of $S$, for a fixed pair $(V', Q')$. This behavior indicates that the piecewise linear approximation made in the $V$ and $Q$ dimensions would not be appropriate. Instead, our strategy is to use a secant approximation in the $S$ dimension at each pair $(V, Q)$ of the same grid. This secant approximation minimizes the average square error between the HPF and the AHPF for $m$ points in the segment between $(V, Q, 0)$ and $(V, Q, S_{ref})$, as shown in expressions (9a), (9b) and in Fig. 9

$$GH_i^k(V,Q,S) = GH_i^{k\prime}(V,Q) + \gamma_S^{i,k} S, \quad k = 1, \ldots, K_i \quad (9a)$$

where

$$\gamma_S^{i,k} = \arg\min_{\gamma} \left[ \frac{1}{m} \sum_{j=1}^{m} \left( HPF(V,Q,S_j) - \left( GH_i^{k\prime}(V,Q) + \gamma S_j \right) \right)^2 \right] \quad (9b)$$

and $S_j = (j/m) S_{ref}$.

The value $S_{ref}$ is based on the plant spillage historical data, and it should not be too high, as more accurate representations are preferred for small values of $S$. Although good approximations would be desired, if possible, for huge amounts of spillage, in these situations water value tends to be near zero, and a higher error would not have much economic impact in subsequent calculations.

We summarize in Fig. 10 the steps to build the AHPF of a given hydro plant.

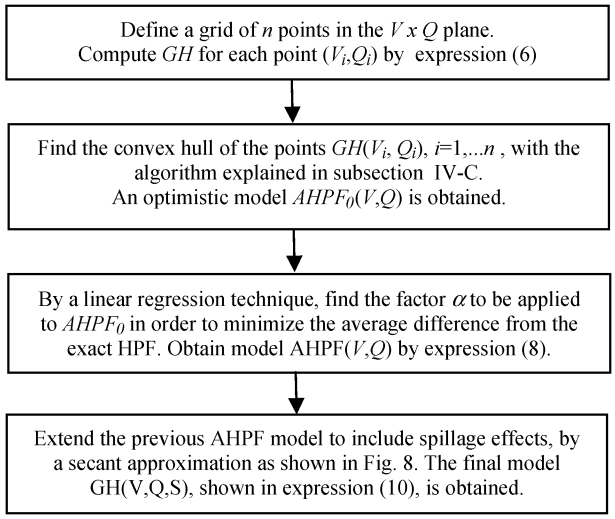



Fig. 10. Flowchart of the procedures to build the 4-D model for hydro generation as a function of storage, turbined outflow, and spillage.

### H. Mathematical Formulation

According to the procedures described in the previous subsections, the mathematical formulation of the AHPF model to be used in the short-term hydrothermal dispatch problem is as follows:

$$\begin{aligned} & GH_i^t \le \alpha \left( \gamma_0^{i,k} + \gamma_V^{i,k} V_i^t + \gamma_Q^{i,k} Q_i^t \right) + \gamma_S^{i,k} S_i^t \\ & \underline{GH_i} \le GH_i^t \le \overline{GH_i}, \quad \underline{V_i} \le V_i^t \le \overline{V_i}, \quad \underline{Q_i} \le Q_i^t \le \overline{Q_i} \\ & \underline{S_i} \le S_i^t \le \overline{S_i}, \quad i = 1, \ldots, NH, t = 1, \ldots, T \\ & k = 1, \ldots, K_i \end{aligned} \quad (10)$$

where $NH$ is the number of hydro plants, $T$ is the number of time steps, and the limits $\underline{x}$ and $\overline{x}$ for each variable $x$ are given. The parameters for each hyperplane $k$ of plant $i$ are $\gamma_0^{i,k} \ge 0$, $\gamma_V^{i,k} \ge 0$, $\gamma_Q^{i,k} \ge 0$ and $\gamma_S^{i,k} \le 0$.

Note that all the points below the AHPF curve of each hydro plant are feasible for the optimal dispatch problem. However, the search for the minimum cost solution will force the hydro generation to touch the curve, since, for any point strictly below the curve, it is possible to find another point of the curve that either spends less water for the same power output, or generates more power spending the same amount of water.

## V. Numerical Results

The proposed AHPF model was implemented in a model called DESSEM, developed by CEPEL for the daily operation of the Brazilian electrical system [28]. Operation planning in Brazil is still done in a centralized way, in a cost-minimization basis [29]. So, the Brazilian ISO has to determine, each day, the one week ahead dispatch of hydro and thermal plants.

DESSEM Model solves the short-term hydrothermal dispatch problem by the so called dual dynamic programming technique, which uses a multistage Benders decomposition approach [30]. It assumes a predetermined hydro and thermal unit commitment along the week. The HPF is then considered for those hours where at least one hydro unit of the plant is committed, and the

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

TABLE III
SENSITIVITY ANALYSIS FOR DISCHARGE DISCRETIZATION

| Number of points for $Q$ | Average AHFP error (%) | Optimal solution ($10^6$\$) | time (min:sec) |
|---|---|---|---|
| 1 (const. head) | 2.07 | 32,645.43 | 01:13 |
| 5 | 0.18 | 32,645.35 | 01:31 |
| 10 | 0.12 | 32,645.35 | 01:58 |
| 15 | 0.11 | 32,645.37 | 02:19 |
| 25 | 0.10 | 32,645.37 | 03:20 |

number of units operating is taken into account by properly adjusting the $Q$ and $GH$ limits for each hour in (10). The program is written in FORTRAN and uses OSL [31] as the solving library for the linear subproblems. At the end of the week a system cost-to-go function calculated by the medium term model in the CEPEL optimization chain [29] was considered, in order to provide a value for the water stored in the reservoirs.

The study consists in the day-ahead network constrained hydrothermal dispatch problem for the Brazilian system, composed of 110 hydro plants, 59 thermal plants, 3457 buses and 4889 transmission lines. The electrical network is represented as a dc model, with line flow limits constraints. Thermal plants modeling includes generation limits, ramp rate constraints, and piecewise linear production costs. Startup/shutdown costs, fuel switching, and fuel consumption/emission constraints were not considered. The application of the hydro plants production function model presented in this paper to a problem with such characteristics is straightforward, due to its piecewise linear formulation. There are four pumped-storage hydro plants, whose pumping modes are modeled as separated units.

The period of study is composed of 168 h, and the average system demand is about 43 000 MW. Several electrical and hydraulic constraints are included. The data for the study is from the third week of April 2006, and is too lengthy to list in the paper. It can be obtained at the Brazilian ISO web site: http://www.ons.org.br.

The AHPF model was implemented in all hydro plants of the study case. To assess the accuracy of our approximation, we calculated, for each hour and each hydro plant, the exact generation associated with the hydraulic operation $(V, Q, S)$ obtained by the model. These generation values, obtained by the analytical expression (6), were compared with the ones obtained by the AHPF model. All tests were run on a Pentium 4 3-GHz 504-MB RAM computer.

### A. Sensitivity Analysis in the Turbined Outflow Representation

The first analysis aims to verify the accuracy of the approximation as the discretization of $Q$ is improved. The effect of the storage is represented by a linear approximation around the known initial volume $V_0$ for each reservoir. We have used five, ten, 15 and 25 grid points for $Q$, uniformly distributed between $Q = \underline{Q}$ and $Q = \overline{Q}$.

From results reported in Table III, we verify that the linear approximation employed for the storage level was enough to keep average deviations between AHPF and HPF very small, provided that we use at least five points for $Q$. We conclude that the effect of the stored volume on the hydro generation is actually almost linear in the short-term planning. However, higher deviations may occur as we increase the horizon of study to comprise several weeks and if we face scenarios with very high inflows to the reservoirs, as reported in [27].

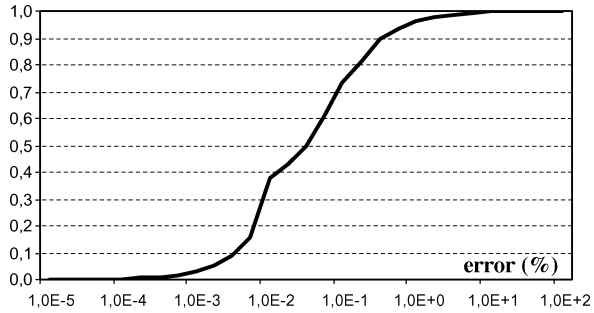



Fig. 11. Distribution of errors in the proposed AHPF model.

The first row shows the results considering a constant water head. We conclude that the use of a piecewise linear model for the HPF does not increase significantly the CPU time.

The average AHPF error, shown in the second column, is calculated by the following expression:

$$\frac{1}{NH \times T} \sum_{t=1}^{T} \sum_{i=1}^{NH} \left[ 100 \left( \frac{GH_i^t - HPF_i\left(V_i^t, Q_i^t, S_i^t\right)}{HPF_i\left(V_i^t, Q_i^t, S_i^t\right)} \right) \right] \tag{11}$$

where $GH_i^t$, $V_i^t$, $Q_i^t$ and $S_i^t$ are outputs of the model, and $HPF(.)$ is the exact hydro production function, calculated *a posteriori* by expression (6).

We note that a discretization of five points for the turbined outflow is quite satisfactory, yielding an average error around 0.18%. However, improvements are observed with acceptable increases in computational times for a discretization of up to ten points. Over ten points, the small improvement in the approximation does not compensate the increase in CPU time. We conclude that there is a maximum number of grid points for $Q$ for which significant improvement in the modeling of the HPF can be obtained.

Table III shows only average errors. Fig. 11 shows the cumulative distribution of these errors. Note that a logarithmic scale was used.

The distribution is quite asymmetric, with more than 90% of errors lower than 1.0%. The biggest errors—between 10% and 40%—occurred for very small values of generation, where small absolute differences lead to high relative deviations. We conclude that our model was able to represent the hydro plants' generation function quite accurately.

In view of the good results obtained, the next analysis considers for all plants a discretization grid of five points in $Q$.

### B. Sensitivity Analysis in the Storage Representation

We now detail the results for some hydro plants whose reservoirs are small and can have a large relative head variation during a short time horizon. The basic data for these plants are shown in Table IV.

We have included severe final storage target constraints for these reservoirs to force a large storage variation during the week. Table V shows the average AHFP error for each plant,

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
MAIN DATA FOR THE SMALL HYDRO PLANTS DETAILED

| Plant | Power Capacity (MW) | # units | Storage capacity ($hm^3$) | Long term productivity ($MW/(m^3/s)$ |
|---|---|---|---|---|
| Camargos | 46 | 2 | 672 | 0.21 |
| Corumba I | 375 | 3 | 1030 | 0.66 |
| Funil | 222 | 3 | 605 | 0.57 |
| Miranda | 408 | 3 | 146 | 0.61 |
| P. Cavalo | 160 | 2 | 880 | 0.88 |
| P. Estrela | 112 | 2 | 44 | 0.45 |

TABLE V
SENSITIVITY ANALYSIS FOR STORAGE DISCRETIZATION—AVERAGE AHFP ERROR (%)

| Plant | Initial Storage (%) | Final Storage (%) | $\Delta V$ used for the window width | | | | |
|---|---|---|---|---|---|---|---|
| | | | 0% | 1% | 5% | 10% | 50% |
| Camargos | 52.8 | 24.0 | 0.61 | 0.47 | **0.46** | 0.91 | 5.57 |
| Corumba I | 37.0 | 11.0 | 1.23 | 1.81 | 1.34 | **0.86** | 2.74 |
| Funil | 32.9 | 8.0 | 1.01 | 1.19 | 0.77 | **0.66** | 3.17 |
| Miranda | 71.5 | 10.0 | 0.28 | **0.00** | 0.01 | 0.01 | 0.04 |
| P. Cavalo | 33.9 | 6.0 | 0.23 | **0.00** | 0.00 | 0.00 | 0.01 |
| P. Estrela | 95.0 | 0.0 | 7.96 | 10.81 | 9.68 | 7.64 | **3.72** |
| **All Plants** | - | - | **0.33** | **0.35** | **0.34** | **0.31** | **0.37** |

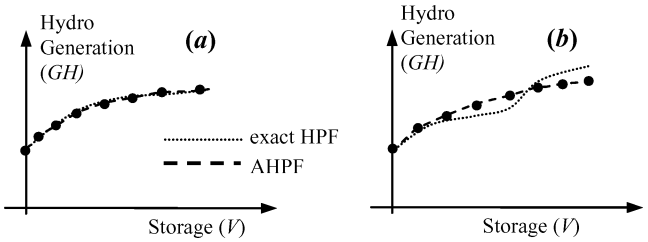



Fig. 12. Quality of approximation as the number of points is increased for the concave (*a*) and nonconcave (*b*) case, for a fixed value of $Q$.

as the discretization window for the storage $(V_0 - \Delta V/2, V_0 + \Delta V/2)$ varies from a width $(\Delta V)$ of 0% (only one grid point for $V$) to 100% of the maximum storage. We marked in bold type the smallest errors for each plant.

In general, higher differences between the AHPF model and the exact HPF correspond to higher variations of storage of a hydro plant during the planning horizon. However, magnitudes of the errors are highly dependent on the shape of the real hydro production function HPF. We consider two extreme situations.

*1) Real HPF Is Concave:* In this case, we can make deviations between the HPF and AHPF arbitrarily small, by increasing the number of points in the discretization, see graph (*a*) in Fig. 12. Examples of this case are Miranda and P.Cavalo power plants.

*2) Real HPF Is Nonconcave:* In this case, since we are approximating a nonconcave curve by a concave piecewise linear model, there is a minimum average error that cannot be avoided, even if the number of points is increased, see graph (*b*) in Fig. 12. An example of this case is P. Estrela hydro plant.

Since the HPF of a hydro plant is in general composed of a mix of both nonconcave and concave subregions, the quality of the approximation will depend on the proportion in which these two situations occur and also on the operation point $(V, Q)$ of the plant.

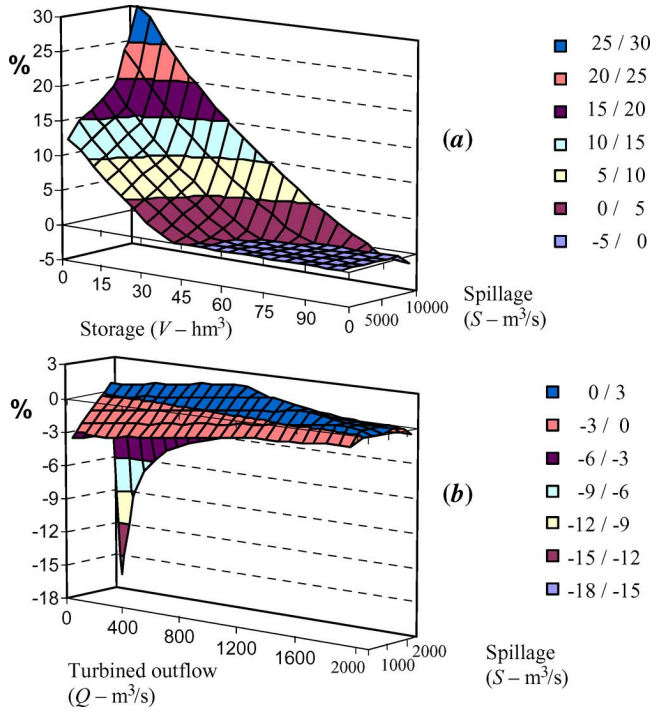



Fig. 13. Differences between the AHPF and HPF functions with spillage—Salto Santiago hydro plant.

## C. Representation of the Spillage Effect

We illustrate from Figs. 13–15 the accuracy in the representation of spillage effects in the AHPF model for three important hydro plants of the Brazilian system. In graphs (a), we assigned to the turbined outflow $Q$ its average value along the week and show the relative differences between the HPF and the AHPF as both $V$ and $S$ vary simultaneously. In graphs (b), we assigned to $V$ the initial storage in the reservoir, and show the differences between these two functions as both $Q$ and $S$ vary.

For Salto Santiago and Tucuruí hydro plants, high deviations are observed in charts (*a*) for combinations of small values of storage and high values of spillage. In charts (*b*), high deviations usually occur for small values of turbined outflow. In both situations, deviations are due to the nonconvexity of the region below the real hydro production function in the dimension of $S$. However, these situations where the highest deviations appeared are rarely seen in practice in the system operation. Actually, spillage often occurs when the storage of the reservoir is close to its capacity, and in this case, the water value tends to be near zero. For this reason, deviations of the AHPF function for higher values of spillage do not have significant economical impact in the system cost minimization.

Among all plants in the Brazilian system, Tucuruí is the one with the largest average errors for the spillage modeling in the HPF. Even for this plant, we can see that the highest deviation that can occur is around 15%.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

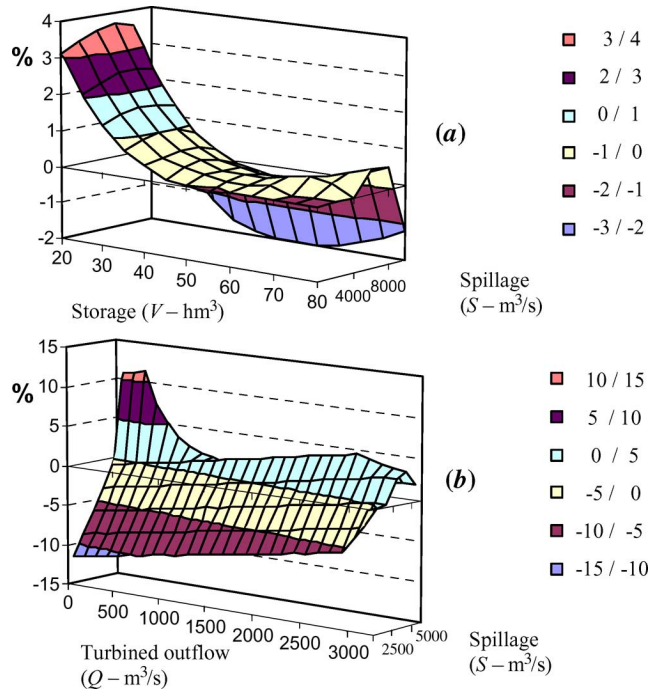



Fig. 14. Differences between the AHPF and HPF functions with spillage—Marimbondo hydro plant.

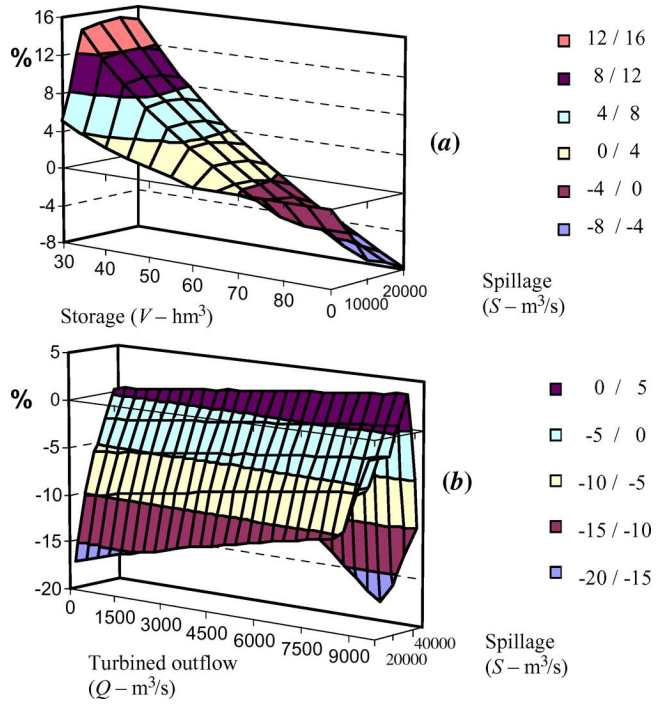



Fig. 15. Differences between the AHPF and HPF functions with spillage—Tucuruí hydro plant.

### D. Dispatch Results

We show now dispatch results for some hydro plants along the day/week. Two situations can be distinguished.

*1) Spillage Did Not Occur:* In this case, deviations are likely to be very small, except in the case the HPF is highly nonconcave, as discussed in Section V-B. Table VI shows the average weekly generation given by both AHPF and HPF functions, for the three plants analyzed in Section V-C. The deviation for Salto Santiago was null because it reached its maximum generation capacity of 1420 MW.

TABLE VI
AVERAGE WEEKLY GENERATION BY THE REAL HPF AND THE AHPF MODEL

| Plant | HPF (MW) | AHPF (MW) | deviation (%) |
|---|---|---|---|
| S. Santiago | 1420.00 | 1420.00 | 0.00% |
| Marimbondo | 599.48 | 600.36 | + 0.15% |
| Tucuruí | 1920.92 | 1920.00 | – 0.05% |

TABLE VII
GENERATION IN SOME HOURS OF THE FIRST DAY, BY THE REAL HPF AND THE AHPF MODEL—MARIMBONDO POWER PLANT

| Hour | Load level | HPF (MW) | AHPF (MW) | deviation (%) |
|---|---|---|---|---|
| 1 | low | 330.00 | 329.96 | – 0.01% |
| 15 | Medium | 603.25 | 601.76 | – 0.25% |
| 19 | peak | 977.75 | 976.53 | – 0.13% |

TABLE VIII
HIGHEST DEVIATIONS OF THE AHPF MODEL WHEN SPILLAGE OCCURRED

| Plant | Hour | HPF (MW) | AHPF (MW) | deviation (%) |
|---|---|---|---|---|
| P. Passos | 3 | 43.49 | 46.10 | + 5.99 |
| Corumba | 1 | 45.66 | 46.39 | + 1.59 |
| S. Caxias | 1 | 162.64 | 159.05 | - 2.21 |

Table VII shows, for Marimbondo power plant, the highest deviations at each load level for the first day.

*2) Spillage Occurred:* In this second situation, deviations may be higher, as discussed in Section V-C. Table VIII shows the highest deviations verified in our study for the first day, in situations where spillage occurred.

### E. Discussion

Nowadays in the Brazilian power market, the ISO determines weekly generation targets for hydro and thermal plants considering a forecast unit commitment declared by the GENCOs at the beginning of the week. Results presented form Sections V-A–C suggest that the proposed model for hydro generation is quite reasonable to be used in a program that would determine the day-ahead hydrothermal system dispatch in an hourly step, based on more refined unit commitment data. However, in the real time system operation, changes in the commitment of the units of a hydro plant would affect its real HPF and might increase the deviations from the AHPF used to compute the plant dispatch. In such situations, real time adjustments in the generation of the hydro plants would be necessary in order to avoid operation in low efficiency levels.

## VI. CONCLUSIONS

This paper presented a four-dimensional piecewise linear model for the power production function of hydro plants, to be used in algorithms for solving optimal hydrothermal dispatch

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

problems. The advantage of the model is to consider in a single function the main elements that cause head variation in hydro plants: storage, turbined outflow and spillage. A real study case with the Brazilian system, considering hydraulic and electrical network constraints, was performed. The small relative deviations between the real hydro production function and the production given by the model confirm the accuracy of the approximation. Low CPU times show the applicability of the proposed approach to the day ahead network constrained short-term hydrothermal dispatch problem for large-scale systems. This model can also be applied to the hydrothermal scheduling problem including thermal unit commitment constraints.

Improvements to this work are being developed, and include: 1) a nonuniform discretization in order to better represent certain ranges of generation; 2) a dynamic adjustment of the model according to the results that are obtained during the dispatch problem solving procedure. This second improvement is important to avoid that the breakpoints of the approximated hydro production function are fixed a priori, as the optimal solution in a simplex-based algorithm tends to reach these points. We also intend to compare the results of the aggregated model, shown in this paper, with the results that would be obtained if an individual curve was considered for each hydro unit of the plant.

## ACKNOWLEDGMENT

The authors wish to acknowledge, in memoriam, the many important contributions of Mr. L. A. Terry during the development of this work. The authors would also like to thank Mrs. C. Sagastizábal for her thorough review of the article, and the anonymous reviewers for the contributions given.

## REFERENCES

[1] N. V. Arvantidis and J. Rosing, "Composite representation of multireservoir hydroelectric power system," *IEEE Trans. Power App. Syst.*, vol. PAS-89, no. 2, pp. 319–326, Feb. 1970.

[2] M. E. P. Maceira, C. B. Mercio, and B. G. Gorestin, "Energy evaluation of the north/northeastern and south/southeastern interconnection with NEWAVE model," in *Proc. VI Symp. Specialists in Electric Operational and Expansion Planning—SEPOPE* (in Portuguese), Salvador, Brazil, May 1998.

[3] A. J. Conejo, J. M. Arroyo, J. Contreras, and F. A. Villamor, "Self-scheduling of a hydro producer in a pool-based electricity market," *IEEE Trans. Power Syst.*, vol. 17, no. 4, pp. 1265–1272, Nov. 2002.

[4] A. Arce, T. Ohishi, and S. Soares, "Optimal dispatch of generating units of the Itaipú hydroelectric plant," *IEEE Trans. Power Syst.*, vol. 17, no. 1, pp. 154–158, Feb. 2002.

[5] E. C. Finardi and E. L. da Silva, "Solving the hydro unit commitment problem via dual decomposition and sequential quadratic programming," *IEEE Trans. Power Syst.*, vol. 21, no. 2, pp. 835–844, May 2006.

[6] C. Li, E. Hsu, A. J. Svoboda, C. Tseng, and R. B. Johnson, "Hydro unit commitment in hydro-thermal optimization," *IEEE Trans. Power Syst.*, vol. 12, no. 2, pp. 764–769, May 1997.

[7] S. Soares and C. T. Salmazo, "Minimum loss predispatch model for hydroelectric power systems," *IEEE Trans. Power Syst.*, vol. 12, no. 3, pp. 1220–1228, Aug. 1997.

[8] X. Guan, A. J. Svoboda, and C. Li, "Scheduling hydro power systems with restricted operating zones and discharge ramping constraints," *IEEE Trans. Power Syst.*, vol. 14, no. 1, pp. 126–131, Feb. 1999.

[9] F. J. Heredia and N. Nabona, "Optimum short-term hydrothermal scheduling with spinning reserve through network flows," *IEEE Trans. Power Syst.*, vol. 10, no. 3, pp. 1642–1651, Aug. 1995.

[10] A. Johannesen, A. Gjelsvik, O. B. Fosso, and N. Flatabo, "Optimal short term hydro scheduling including security constraints," *IEEE Trans. Power Syst.*, vol. 6, no. 2, pp. 576–583, May. 1991.

[11] R. A. Ponrajah and F. D. Galiana, "Systems to optimize conversion efficiencies at Ontario Hydro's hydroelectric plants," *IEEE Trans. Power Syst.*, vol. 13, no. 3, pp. 1044–1050, Aug. 1998.

[12] O. Nilsson and D. Sjelvgren, "Variable splitting applied to modelling of start-up costs in short term hydro generation scheduling," *IEEE Trans. Power Syst.*, vol. 12, no. 2, pp. 770–775, May 1997.

[13] E. Ni, X. Guan, and R. Li, "Scheduling hydrothermal power systems with cascaded and head-dependent reservoirs," *IEEE Trans. Power Syst.*, vol. 14, no. 3, pp. 1127–1132, Aug. 1999.

[14] L. Dubost, R. Gonzalez, and C. Lemaréchal, A primal proximal heuristic applied to the unit commitment problem, INRIA, Rapport de recherche no 4978, Oct. 2003.

[15] G. W. Chang, M. Aganagic, J. G. Waight, and J. Medina *et al.*, "Experiences with mixed integer linear programming based approaches on short term hydro scheduling," *IEEE Trans. Power Syst.*, vol. 16, no. 4, pp. 743–749, Nov. 2001.

[16] M. S. Salam, K. M. Nor, and A. R. Hamdan, "Hydrothermal scheduling based Lagrangian relaxation approach to hydrothermal coordination," *IEEE Trans. Power Syst.*, vol. 13, no. 1, pp. 226–235, Feb. 1998.

[17] S. Ruzic and N. Rajakovic, "Optimal distance method for Lagrangian multipliers updating in short term hydro-thermal coordination," *IEEE Trans. Power Syst.*, vol. 13, no. 4, pp. 1439–1444, Nov. 1998.

[18] Z. K. Shawwash, T. K. Siu, and S. O. D. Russel, "The B.C. hydro short term hydro scheduling optimization model," *IEEE Trans. Power Syst.*, vol. 15, no. 3, pp. 1125–1131, Aug. 2000.

[19] H.-C. Chang and P.-H. Chen, "Hydrothermal generation scheduling package: A genetic based approach," *Proc. Inst. Elect. Eng., Gen., Transm., Distrib.*, vol. 145, no. 4, pp. 451–457, Jul. 1998.

[20] L. Bacaud, C. Lemaréchal, A. Renaud, and C. Sagastizábal, "Bundle methods in stochastic optimal power management: A disaggregated approach using preconditioners," *Comput. Optim. Appl.*, vol. 20, pp. 27–244, 2001.

[21] A. S. Belloni, A. L. Diniz, C. A. Sagastizábal, and M. E. Maceira, "Bundle relaxation and primal recovery in unit commitment problems—The Brazilian case," *Ann.Oper. Res.*, vol. 120, pp. 21–44, 2003.

[22] J. G. Gonzalez and G. A. Castro, "Short-term hydro scheduling with cascaded and head-dependent reservoirs based on mixed-integer linear programming," in *Proc. 10th Power Tech Conf.*, Porto, Portugal, Sep. 2001.

[23] S. O. Orero and M. R. Irving, "A genetic algorithm modelling framework and solution technique for short term optimal hydrothermal scheduling," *IEEE Trans. Power Syst.*, vol. 13, no. 2, pp. 501–518, May 1998.

[24] N. Sinha, R. Chakrabarti, and P. K. Chattopadhyay, "Fast evolutionary programming techniques for short-term hydrothermal scheduling," *IEEE Trans. Power Syst.*, vol. 18, no. 1, pp. 214–220, Feb. 2003.

[25] C. E. Zoumas, A. G. Bakirtzis, J. B. Theocharis, and V. Petridis, "A genetic algorithm solution approach to the hydrothermal coordination problem," *IEEE Trans. Power Syst.*, vol. 19, no. 2, pp. 1356–1364, May 2004.

[26] S. H. F. Cunha, S. Prado, and J. P. Costa, "Modelling of the variable productivity of hydro plants based on a hydro production function," in *Proc. XII Simpósio Brasileiro de Recursos Hídricos* (in Portuguese), 1997, pp. 391–397.

[27] A. L. Diniz, M. E. Maceira, and L. A. Terry, "Modelling of the production function of hydro plants in the medium and short term models for interconnected systems planning," in *Proc. IX Symp. Specialists in Electric Operational and Expansion Planning—SEPOPE*, Rio de Janeiro, Brazil, May 2004.

[28] A. L. Diniz, T. N. Santos, and M. E. P. Maceira, "Short term security constrained hydrothermal scheduling considering transmission losses," in *Proc. 2006 IEEE Power Eng. Soc. Transmission Distribution Conf. Expo. Latin America*, Caracas, Venezuela, Aug. 2006.

[29] M. E. P. Maceira, L. A. Terry, and F. S. Costa *et al.*, "Chain of optimization models for setting the energy dispatch and spot price in the Brazilian system," in *Proc. Power System Computation Conf.—PSCC'02*, Seville, Spain, Jun. 2002.

[30] M. V. F. Pereira and L. M. V. G. Pinto, "Multi stage stochastic optimization applied to energy planning," *Math. Program.*, vol. 52, pp. 359–375, 1991.

[31] IBM Optimization Subroutine Library (OSL)—Guide and Reference, Release 2.1, 5th ed. , 1995.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.

**Andre Luiz Diniz** (M'06) received the B.Sc. degree in civil engineering and the M.Sc. and D.Sc. degrees in optimization from the Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil, in 1996, 2000, and 2007, respectively.

Since 1998, he has been a researcher at CEPEL, the Brazilian Electrical Power Research Center, where he has been working in mathematical models for the short and midterm hydrothermal scheduling, including hydro and thermal unit commitment. He is also an Assistant Professor at the Institute of Mathematics and Statistics at the State University of Rio de Janeiro (UERJ).

**Maria Elvira Piñeiro Maceira** received the B.Sc., M.Sc., and D.Sc. degrees in civil engineering from the Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil, in 1983, 1989, and 1994, respectively.

Since 1985, she has been a researcher at CEPEL, where she coordinates the Department of Energy Optimization and Environment. She has been working in mathematical models for the operation and expansion planning of hydrothermal systems, auto-regressive hydrological models, portfolio optimization, and risk analysis. She participated in several research groups in Brazil related to energy studies. She is also an Assistant Professor at the Institute of Mathematics and Statistics at the State University of Rio de Janeiro (UERJ).

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on October 28,2026 at 13:48:56 UTC from IEEE Xplore. Restrictions apply.